

Tracing the Origin: Source Attribution of Social Media Profile Images Using Deep Ensemble Learning

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Abstract—Social media platforms routinely alter image compression, reduce resolution, and strip metadata, severely degrading the signals relied upon for traditional digital forensics. To address this challenge, this paper proposes a platform-aware framework for identifying the provenance of profile images across three source classes: smartphone-captured, internet-sourced, and AI-generated. Drawing on a provenance documented dataset of over 30,000 verified images, a controlled, single-hop routing framework processes the data through Facebook and X to capture realistic platform-induced transformations. The stability of eight deep learning models is then evaluated under these degraded conditions, culminating in an ensemble based source identification approach that utilizes averaging, weighted averaging, and stacking. The resulting stacked meta-learner demonstrates significant resilience to platform processing. These findings highlight the critical necessity of platform-aware training for reliable operational forensic deployment.

Index Terms—Image forensics, image provenance, source attribution, social media forensics, AI generated images, deep learning, platform induced transformations, Facebook, X (Twitter).

I. INTRODUCTION

Social media profile images play a strong role in how users judge authenticity online, but the same images are also exploited in identity fraud and coordinated inauthentic behavior [1]. In fraudulent accounts, profile images commonly come from three sources: smartphone captures, public internet sources, or artificial intelligence (AI) generation.

Conventional digital image forensics relies on traces such as sensor noise, compression history, metadata patterns, and generative artifacts to infer an image’s origin [2]–[4]. However, platforms such as Facebook and X alter uploaded content through recompression, resizing, and metadata sanitization. These processes can weaken or remove the forensic traces needed for reliable source attribution [3], [5], [6].

This creates a practical gap between laboratory evaluation and real world forensic use, since existing datasets such as FaceForensics++ and DFDC were designed for manipulation and deepfake detection, not still image provenance attribution across smartphone captured, internet sourced, and AI generated images [7], [8]. As a result, classifiers trained only

on clean data may not generalize well to platform processed images or to newer generative domains [4], [9].

To address this gap, this paper proposes a platform aware framework for image provenance classification. By combining provenance tracking with simulated platform degradation, this work makes the following contributions:

- A provenance documented dataset covering AI generated, smartphone captured, and internet sourced images.
- A controlled single hop routing framework simulating realistic transformations across Facebook and X.
- An evaluation of deep learning and ensemble classifiers under platform processed conditions.
- An analysis of classification resilience, highlighting the need for platform aware training in forensic deployment.

II. RELATED WORK

Digital image forensics studies traces left by image creation, editing, and compression. Classical methods detect splicing, resampling, and JPEG inconsistencies, but these cues can be weakened by recompression and resizing [2]. Provenance analysis extends this task to source and transformation history, with prior work showing that social network processing can itself serve as a forensic signal [3].

Social media platforms further complicate attribution by recompressing, resizing, renaming, and stripping metadata from uploaded images [5], [6]. This can make smartphone captured and internet sourced images more difficult to separate after platform processing.

AI generated imagery adds another challenge. GAN and diffusion models can produce realistic synthetic images, while model specific generative fingerprints can support synthetic source attribution [9]. However, detector performance can degrade under domain shift and downstream processing [10].

Deep learning methods, including CNNs, transformers, and pretrained vision models, have become central to image forensic detection because they learn discriminative features directly from data [11]. However, common datasets such as FaceForensics++ and DFDC focus mainly on deepfake or manipulation detection rather than still image provenance

classification across smartphone, internet, and AI generated sources [7], [8]. This work addresses the gap by evaluating platform aware provenance classification on images processed through Facebook and X.

This work differs from prior studies in three ways. First, AI generated image detectors [4], [9], [10] address a binary real versus synthetic task, whereas this work performs three class provenance attribution that also separates two classes of real images by acquisition path. Second, prior social media forensics studies [3], [5], [6] examine platform effects on manipulation detection or single-platform origin classification, not multi-class attribution under controlled routing through multiple platforms. Third, unlike evaluations on pristine benchmarks [7], [8], all results here are reported on platform processed images.

III. METHODOLOGY

This study develops a provenance-aware framework for image source attribution under Facebook and X processing. The workflow combines source image curation, controlled platform routing, preprocessing, model training, and ensemble evaluation.

A. Dataset and Platform Routing

The dataset is organized into three provenance classes: AI generated, smartphone captured, and internet sourced images. The initial collection contained approximately 43,000 AI generated images, 25,000 cleaned smartphone images, and 60,000 internet sourced images. Smartphone images were acquired from known devices (iPhone and Android) with capture metadata logged at acquisition. Internet sourced images were collected from public web sources using a custom web scraper application and were verified through reverse image search. AI generated images were produced with several AI image generation models, including GAN and diffusion models, with the generating model recorded per image. Duplicates and near duplicates were removed using hash-based and perceptual similarity checks, leaving 37,498 verified original images.

The retained images were routed through Facebook and X using a controlled single hop upload and download process. The resulting derivatives capture platform effects such as recompression, resizing, format normalization, and metadata removal. The final dataset composition and split are summarized in Table I.

The dataset was split 70:15:15 into training, validation, and test sets at the level of the original source image: each original and all of its Facebook and X derivatives were assigned wholly to one partition, so no derivative of a training image appears in validation or testing. Stratification within this grouping preserved provenance class, semantic category, and platform condition.

B. Preprocessing and Model Objective

All images underwent file integrity validation, model input normalization, metadata extraction, and quality screening. Corrupted or low quality files were excluded.

TABLE I
DATASET COMPOSITION AND SPLIT SUMMARY

Component	Count
Retained original images	37,498
Facebook derivatives	41,001
X derivatives	34,310
Platform derivatives	75,311
Main dataset	112,809
Validation support images	15,000
Training set	89,466
Validation set	19,171
Test set	19,172
Total dataset	127,809

Image provenance classification is formulated as a three-class problem with $K = 3$. Given an input image x , a model parameterized by θ produces logits $f_k(x; \theta)$, which are converted into class probabilities using softmax:

$$p(y = k | x; \theta) = \frac{\exp(f_k(x; \theta))}{\sum_{j=1}^K \exp(f_j(x; \theta))}. \quad (1)$$

The models are trained using categorical cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log p(y = k | x_i; \theta). \quad (2)$$

C. Ensemble Learning Strategies

To improve stability, three ensemble strategies were evaluated using the top four base models selected by validation accuracy. Let $p^{(m)}(x) \in \mathbb{R}^K$ denote the class probability vector produced by the m th base model for input image x , where $M = 4$ and $K = 3$.

The first two strategies combine base model probabilities as

$$p_{\text{ens}}(x) = \sum_{m=1}^M w_m p^{(m)}(x), \quad \hat{y} = \arg \max_k p_{\text{ens},k}(x), \quad (3)$$

where simple averaging uses uniform weights $w_m = 1/M$, and weighted averaging sets $w_m = a_m / \sum_{j=1}^M a_j$, with a_m the validation accuracy of the m th base model. The third strategy is stacking, in which the base model probability vectors are concatenated into a meta feature vector and classified by a multinomial logistic regression meta-learner:

$$\hat{y} = g\left(\left[p^{(1)}(x), \dots, p^{(M)}(x)\right]; \phi\right), \quad (4)$$

where $g(\cdot)$ is trained on validation set meta features and ϕ denotes its learned parameters, with the test set held out for final evaluation.

IV. EXPERIMENTAL RESULTS

Performance is reported on the held out test set using accuracy, per-class precision, recall, and F1, macro-averaged F1, and the Matthews correlation coefficient (MCC), which is robust to class imbalance.

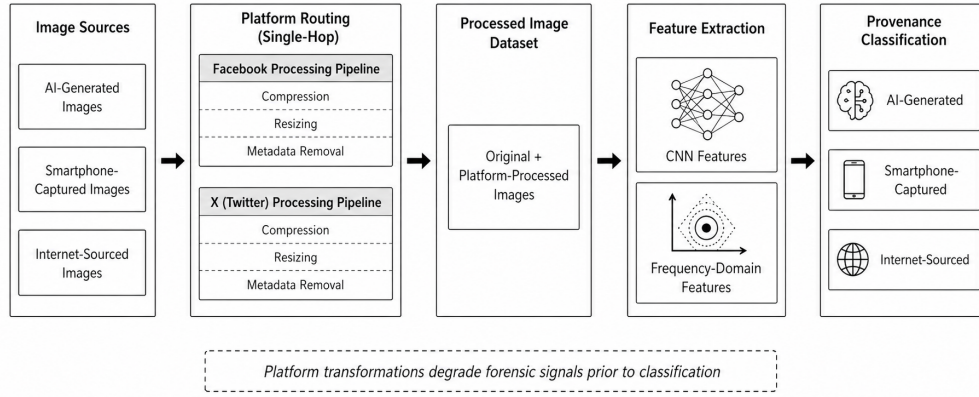


Fig. 1. Overview of the provenance aware pipeline

TABLE II

COMPARISON OF SINGLE AND ENSEMBLE MODELS UNDER FACEBOOK PROCESSING

Model	Type	Acc	F1 (M)	MCC
BEiTv2 Base [12]	Single	0.9329	0.9104	0.8593
ConvNeXtV2 Base [13]	Single	0.9690	0.9570	0.9358
DINOv2 Base [14]	Single	0.9766	0.9677	0.9508
EffNetV2 S [15]	Single	0.9303	0.9076	0.8525
MaxViT Tiny [16]	Single	0.9293	0.9070	0.8525
SigLIP Base [17]	Single	0.9776	0.9696	0.9531
Swin Base [18]	Single	0.9693	0.9573	0.9364
ViT Base [19]	Single	0.9514	0.9343	0.8977
SimpleAvg Top4	Ensemble	0.9815	0.9743	0.9614
WeightedAvg Top4	Ensemble	0.9815	0.9743	0.9614
Stacked MetaLearner	Ensemble	0.9819	0.9749	0.9620

A. Facebook Platform Results

1) *Single Model Performance*: Table II shows the performance of individual models on the Facebook processed test set. Accuracy ranges from 0.93 to 0.98, with SigLIP and DINOv2 achieving the strongest results. The gap between macro and weighted F1 scores indicates class level variation, with smartphone captured images remaining more difficult to classify than AI generated and internet sourced images.

2) *Ensemble Performance*: Table II shows that the stacked meta-learner achieved the best Facebook result, improving over SigLIP Base from 0.9776 to 0.9819 accuracy, 0.9696 to 0.9749 macro F1, and 0.9531 to 0.9620 MCC. The identical SimpleAvg and WeightedAvg results suggest that validation-based weighting did not materially change the ensemble decision boundary.

B. X Platform Results

1) *Single Model Performance*: Table III shows the performance of individual models evaluated on the X-processed test set.

Under X processing, model accuracy ranges from 0.8569 to 0.9316, with SigLIP and DINOv2 achieving the strongest accuracy and macro F1 scores. As with Facebook, transformer-based models show better class level consistency than EfficientNetV2 and MaxViT, although the gap between macro and weighted F1 indicates continued variation across provenance classes.

TABLE III

COMPARISON OF SINGLE AND ENSEMBLE MODELS UNDER X PROCESSING

Model	Type	Acc	F1 (M)	MCC
BEiTv2 Base [12]	Single	0.8627	0.8342	0.7483
ConvNeXtV2 Base [13]	Single	0.9121	0.8901	0.8369
DINOv2 Base [14]	Single	0.9235	0.9023	0.8538
EffNetV2 S [15]	Single	0.8779	0.8476	0.7664
MaxViT Tiny [16]	Single	0.8569	0.8279	0.7409
SigLIP Base [17]	Single	0.9316	0.9126	0.8697
Swin Base [18]	Single	0.9174	0.8971	0.8456
ViT Base [19]	Single	0.9052	0.8831	0.8226
SimpleAvg Top4	Ensemble	0.9311	0.9122	0.8690
WeightedAvg Top4	Ensemble	0.9311	0.9122	0.8690
Stacked MetaLearner	Ensemble	0.9331	0.9132	0.8702

TABLE IV

PER CLASS PERFORMANCE OF THE STACKED ENSEMBLE

Platform	Class	Prec.	Rec.	F1
Facebook	AI generated	0.9729	0.9744	0.9737
Facebook	Smartphone	0.9716	0.9539	0.9626
Facebook	internet sourced	0.9863	0.9906	0.9885
X	AI generated	0.8629	0.8294	0.8458
X	Smartphone	0.9451	0.9439	0.9445
X	internet sourced	0.9449	0.9538	0.9493

2) *Ensemble Performance*: Table III compares single and ensemble models under X processing. SigLIP Base achieved the strongest single-model result with 0.9316 accuracy, 0.9126 macro F1, and 0.8697 MCC, while the stacked meta-learner produced the best overall result with 0.9331 accuracy, 0.9132 macro F1, and 0.8702 MCC.

The gains from ensembling are smaller than under Facebook, suggesting stronger degradation or class overlap under X processing. Still, stacking provides the most stable ensemble strategy, although it does not fully resolve confusion with the internet sourced class.

C. Cross-Platform Comparative Analysis

Table IV reports per-class precision, recall, and F1 for the stacked ensemble, and Table V shows the corresponding confusion matrices.

Table VI reports a file-level check of matched original derivative pairs. Over 95% of derivatives lost all metadata, confirming that both platforms strip EXIF data. Both platforms re-encoded only the internet sourced pair while passing

TABLE V
CONFUSION MATRICES OF THE STACKED ENSEMBLE UNDER FACEBOOK
AND X PROCESSING (ROWS: ACTUAL, COLUMNS: PREDICTED)

Platform	Actual	AI	Internet	Smartphone
Facebook	AI	610	12	4
Facebook	Internet	11	3171	19
Facebook	Smartphone	6	32	786
X	AI	491	100	1
X	Internet	75	2435	43
X	Smartphone	3	42	757

TABLE VI
FILE LEVEL VERIFICATION OF MATCHED ORIGINAL AND DERIVATIVE
PAIRS

Class	Platform	Q_o to Q_d	Size	Verdict
Internet	Facebook	13.81 to 23.08	Same	Re encoded; EXIF stripped
AI	Facebook	9.25 to 9.25	Same	Pixel identical
Smartphone	Facebook	1.70 to 1.70	Same	Byte identical; excluded
Internet	X	13.81 to 15.84	768×1024 to 1080×1080	Re encoded; cropped; upscaled
AI	X	9.25 to 9.25	Same	Pixel identical
Smartphone	X	7.23 to 7.23	Same	Pixel identical; EXIF stripped

smartphone and AI pairs through pixel identical, showing that re-encoding is conditional. Facebook quantized more coarsely but preserved dimensions, while X re-encoded more lightly and resampled the image to 1080×1080 , re-interpolating every pixel. This resampling and re-encoding attenuates the pixel aligned sensor and generative residuals attribution depends on, making it the likely driver of the performance gap; since both platforms strip metadata, the difference is primarily pixel-domain. Platform pipelines are proprietary and undocumented, so the transformations reflect each platform’s behavior at collection time.

Error patterns differ by platform. Under X processing, overall accuracy remains high at 0.9331, but AI recall declines to 0.8294 and confusion between AI and internet images becomes the leading error. This platform dependent degradation supports the central argument that evaluation on pristine data can overstate deployed performance and motivates future work on frequency domain features.

V. CONCLUSION AND FUTURE WORK

This paper presented a platform aware framework for classifying AI generated, smartphone captured, and internet sourced images after Facebook and X processing. The results show that provenance signals remain detectable, although performance varies by platform and class. Transformer based models and stacking produced the strongest results.

Most classification errors involved internet sourced images. Facebook processing, these errors occurred mainly between internet sourced and smartphone images. Under X processing, error rates increased between AI generated and internet sourced images. Future work will examine frequency domain features, DCT statistics, noise residuals, multi hop routing, and additional platforms. Since platform pipelines are proprietary and change without notice, deployed classifiers will require periodic retraining. The dataset was curated from images

collected for this study with participant consent, and such tools should support account integrity and forensic investigation rather than surveillance.

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